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EXCEL ASSIGNMENT

1. What is the difference between 'Paste' and 'Paste Special' in Excel? Briefly explain with examples.

- Paste

Paste copies *everything* from the copied cells:

- Values
- Formulas
- Formatting (colors, borders, font)

Example:
If you copy a cell containing `=A1+B1` with yellow background and paste it, the **formula and yellow color** both will be pasted.

Paste Special

Paste Special lets you choose *what exactly* you want to paste.

Common options with examples:

- **Values** → Pastes only the result (not the formula)
  - `=A1+B1` → pastes `50`
- **Formulas** → Pastes only the formula
- **Formats** → Pastes only cell formatting (color, font, borders)
- **Transpose** → Converts rows into columns (or vice versa)

Example:
Copy a total calculated by formula and use **Paste Special** → **Values** to paste only the number, without the formula.

2.Describe the functions and usefulness of 'Freeze Panes' and 'Split Panes' in Excel?

- **Freeze Panes**

Function:
Freeze Panes keeps selected rows or columns **fixed (visible)** while you scroll through the worksheet.

Usefulness:

- Helps you view headings while scrolling large data
- Prevents losing track of row/column labels

Example:
Freezing the **top row** keeps column headings visible when scrolling down a long list of data.

Split Panes

Function:
Split Panes divides the worksheet window into **multiple sections** that can be scrolled **independently**.

Usefulness:

- Compare different parts of the same worksheet at once
- View data from different areas simultaneously

Example:
Splitting the sheet allows you to view sales data at the top and bottom of the sheet at the same time.

3. Explain the difference between inserting a new row and inserting a new column in Excel. Can you insert multiple rows or columns at once?

- **Inserting a New Row**
- Adds a **horizontal** line of cells.
- The new row is inserted **above** the selected row.
- Existing rows move **downward**.

Example:
Selecting row 5 and inserting a row will add a new row **above row 5**.

Inserting a New Column

- Adds a **vertical** line of cells.
- The new column is inserted **to the left** of the selected column.
- Existing columns move **to the right**.

Example:
Selecting column C and inserting a column will add a new column **before column C**.

- **Multiple insert** → Yes, by selecting multiple rows/columns first.

4. What are logical functions in Excel? Provide examples of at least two logical functions and their applications.

- **Logical Functions in Excel**
- **Logical functions** are used to **test conditions** and return results based on whether the condition is **TRUE or FALSE**.
- ---
- **Examples of Logical Functions**
- **1. IF Function**
- **Purpose:**
  Checks a condition and returns one value if TRUE and another if FALSE.
- **Example:**

- `=IF(A1>=50,"Pass","Fail")`
-
- **Application:**
  Used to calculate **pass/fail**, **bonus eligibility**, or **grades**.
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- **2. AND Function**
- **Purpose:**
  Returns TRUE only if **all conditions** are true.
- **Example:**
- `=AND(A1>=50, B1>=40)`
-
- **Application:**
  Used when **multiple conditions** must be satisfied, such as attendance **and** marks criteria.
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- **(Optional) Another Common Logical Function**
- **OR Function**
- **Purpose:**
  Returns TRUE if **any one** condition is true.
- **Example:**
- `=OR(A1="Yes", B1="Yes")`
-
- **Application:**
  Used for **eligibility checks** where one condition is sufficient.

5. Discuss the purpose of 'XLOOKUP' and how it differs from the traditional 'VLOOKUP' function.

- The `XLOOKUP` function is a modern, versatile replacement for traditional Excel lookup functions like `VLOOKUP`, `HLOOKUP`, and `INDEX/MATCH`. Introduced in Microsoft 365 and Excel 2021, its primary purpose is to find a specific value in a range or array and return a corresponding value from another range, overcoming the significant, structural limitations of the older `VLOOKUP` function.

Feature	VLOOKUP (Legacy)	XLOOKUP (Modern)

Lookup Direction	Left-to-right only (leftmost column must hold key).	Any direction (left or right, top or bottom).
Default Match Mode	Approximate match (requires `FALSE` for exact).	Exact match (safer default).
Column Structure	Requires a hardcoded column index number.	References specific ranges, avoiding broken formulas.
Error Handling	Needs extra `IFERROR` formula.	Built-in `if_not_found` argument.
Return Values	Single column only.	Can return multiple columns (spilling).
Data Sorting	Requires sorting for approximate match.	Works on unsorted data.
Compatibility	All Excel versions.	Excel 365, 2021+.

6. Create a worksheet titled 'Employee Data' with columns: Name, Age, Department. Add 5 rows of data. Format as

follows: ● Bold and center-align the header row ● Apply a fill color ● Auto-fit column width?

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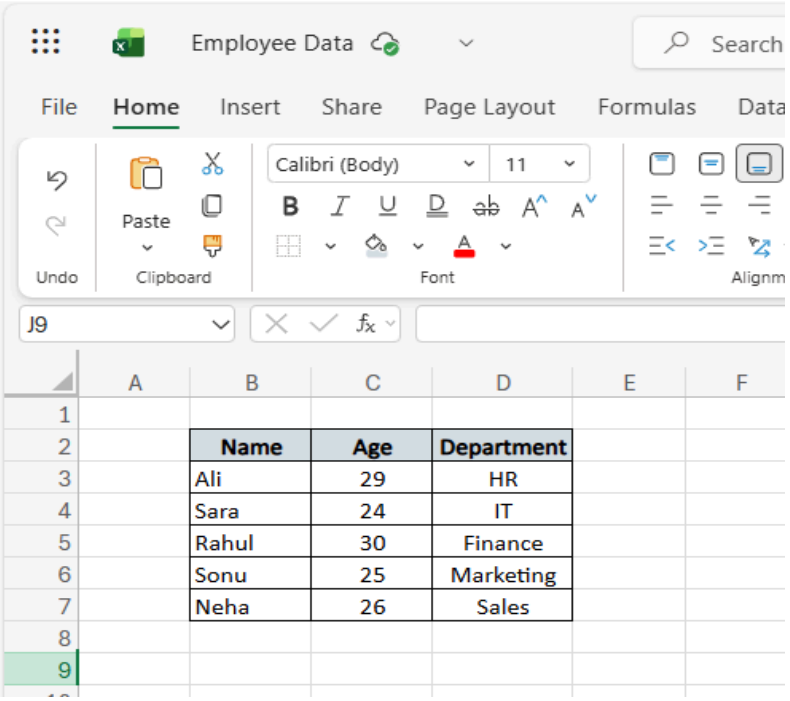

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7. Demonstrate how to insert and delete multiple rows and columns in Excel.

- To insert and delete multiple rows.
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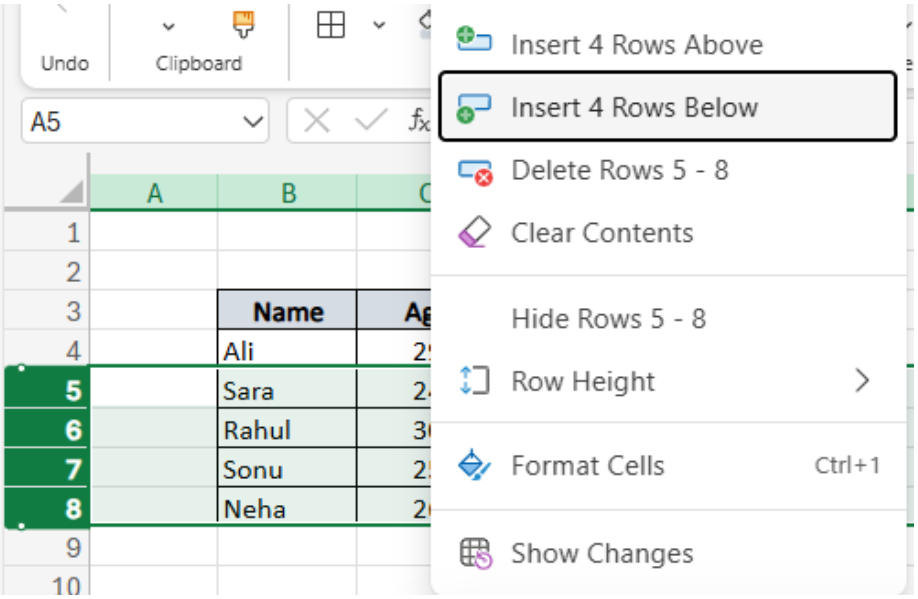

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- To insert and delete multiple columns.
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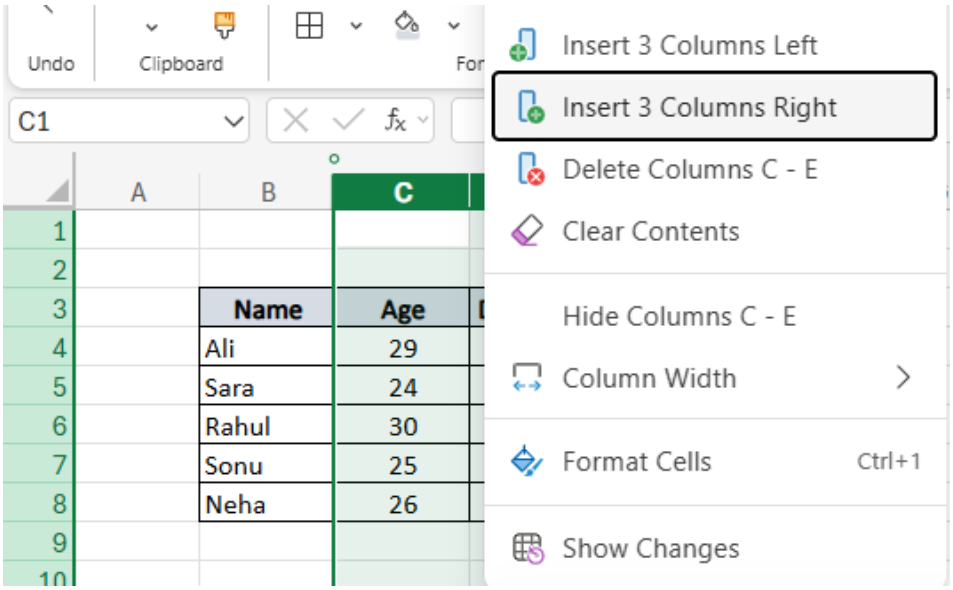

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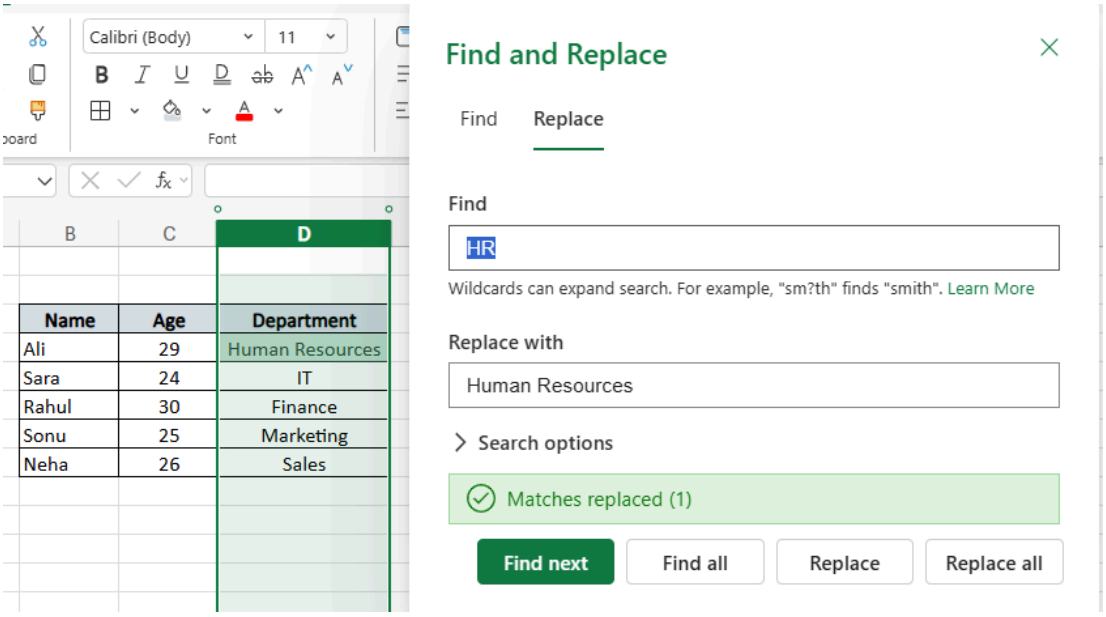
8. Use Excel's 'Find and Replace' feature to update department names in a sample table.



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9. Create a small numerical dataset and apply the following functions: • AVERAGE • MAX • MIN.

- AVERAGE.

=AVERAGE(B3:B7)

B	C	D	E
Number	Result		
10	30		
20			
30			
40			
50			

- MAX

=MAX(B3:B7)

B	C	D
Number	Result	
10	30	
20	50	
30		
40		
50		

- MIN

=MIN(B3:B7)

B	C	D
Number	Result	
10	30	
20	50	
30	10	
40		
50		

10. You're working with a dataset that contains missing values. As a Data Scientist, explain how you'd detect and handle missing data using Excel. Mention tools like: ● Go To Special ● ISBLANK ● COUNTBLANK

- ISBLANK

=ISBLANK(D5)

Name	Age	Department
Ali	29	HR
Sara		IT
Rahul	30	
Sonu		Marketing
Neha	26	

TRUE

- COUNTBLANK

=COUNTBLANK(C2:D7)

Name	Age	Department
Ali	29	HR
Sara		IT
Rahul	30	
Sonu		Marketing
Neha	26	

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